

Unit 1

8 warm-ups · 32 vocabulary terms

By the end of this unit, I can...

- ☐ I can find a point's (x, y) on the CMU canvas.
- ☐ I can explain why y increases going down.
- ☐ I can write `Circle(cx, cy, r)` for a circle I see.
- ☐ I can write `Rect(left, top, w, h)` for a rectangle.
- ☐ I can name the shape functions `Oval`, `Line`, and `Star`.
- ☐ I can read the arguments a `Star` needs.
- ☐ I can set a shape's fill color.
- ☐ I can add a border and set its width.
- ☐ I can predict which shape is drawn on top.
- ☐ I can use opacity to see through a shape.
- ☐ I can put text on the canvas with `Label`.
- ☐ I can position and size a `Label`.
- ☐ I can rotate a shape with `rotateAngle`.
- ☐ I can describe a rotation in degrees.
- ☐ I can read a multi-shape drawing and list its shapes.
- ☐ I can plan a small scene before I code it.

How to use these pages

Each warm-up is two pages. Page 1 is the header, the START HERE strip (A–E), and Vocabulary. Page 2 is Part 1 (everyone) and Part 2 (keep going if you finish early). Mark your own or a partner's work — it does not count against you.

Name	Date	Period
Partner	Points ____ / ____	Marked by

Warm-up 1 — The CMU coordinate plane

START HERE

A	Pace check — circle one: () feeling good () a bit distracted () stuck — I could use help
B	☐ got it ☐ shaky ☐ not yet I can find a point's (x, y) on the CMU canvas.
C	☐ got it ☐ shaky ☐ not yet I can explain why y increases going down.
D	Looking back at last class — one thing that clicked, and one thing still fuzzy:
E	Something you are looking forward to this week:

Vocabulary — write each in your own words

coordinate plane

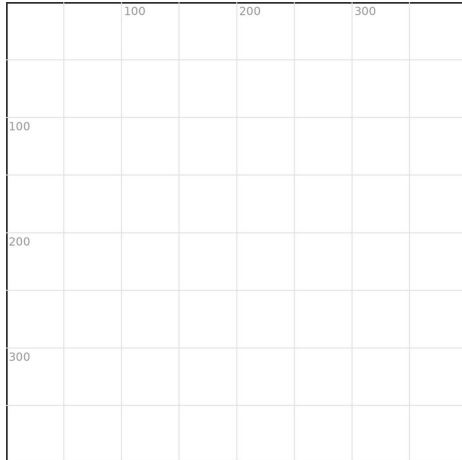
origin

x-coordinate

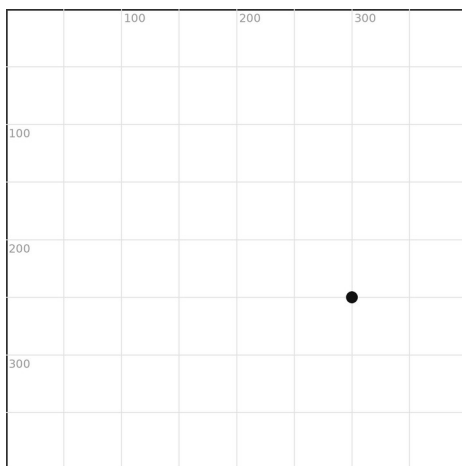
y-coordinate

Part 1 — core work

The origin (0, 0) is the TOP-LEFT corner of the CMU canvas. On the grid, mark the point (200, 100) with a dot and label it.



Write the (x, y) coordinates of the dot drawn below.



In CMU graphics y counts DOWN. Which point is LOWER on the screen: (100, 50) or (100, 350)? Explain in one sentence.

PART 2 — keep going if you finish early.

Name the corner each point lands in: (0, 400), (400, 0), (400, 400).

Name	Date	Period
Partner	Points ____ / ____	Marked by

Warm-up 2 — Circle and Rect

START HERE

A	Pace check — circle one: () feeling good () a bit distracted () stuck — I could use help
B	☐ got it ☐ shaky ☐ not yet I can write Circle(cx, cy, r) for a circle I see.
C	☐ got it ☐ shaky ☐ not yet I can write Rect(left, top, w, h) for a rectangle.
D	Looking back at last class — one thing that clicked, and one thing still fuzzy:
E	A person you could ask for help if you get stuck:

Vocabulary — write each in your own words

Circle

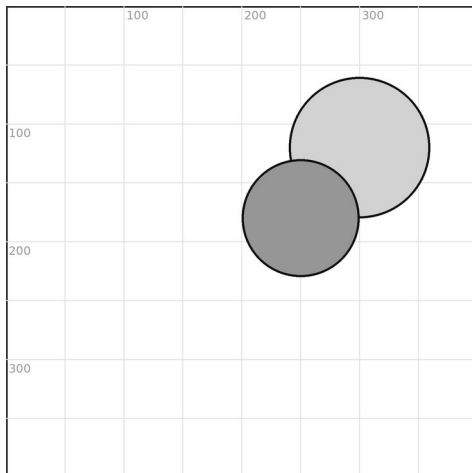
Rect

radius

argument

Part 1 — core work

Two circles are drawn below.



```
Circle(centerX, centerY, radius)
```

Write the CMU code for the SMALLER circle using the pattern above.

A Rect starts at its TOP-LEFT corner: Rect(left, top, width, height). Write the code for a box 150 wide and 80 tall whose top-left is (60, 260).

PART 2 — keep going if you finish early.

Change the big circle's radius to 90. Does it grow toward the center or toward the edge? Explain.

Name	Date	Period
Partner	Points ____ / ____	Marked by

Warm-up 3 — Ovals, lines, and stars

START HERE

A	Pace check — circle one: () feeling good () a bit distracted () stuck — I could use help
B	☐ got it ☐ shaky ☐ not yet I can name the shape functions Oval, Line, and Star.
C	☐ got it ☐ shaky ☐ not yet I can read the arguments a Star needs.
D	Looking back at last class — one thing that clicked, and one thing still fuzzy:
E	One thing that is going well outside of this class:

Vocabulary — write each in your own words

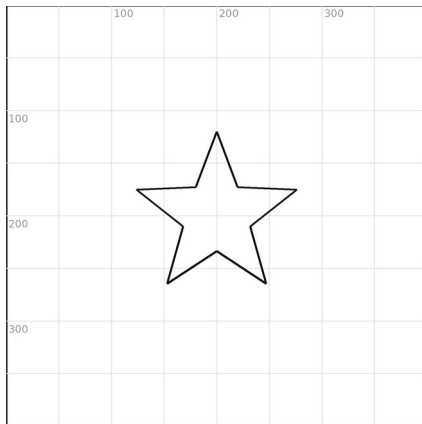
Oval

Line

Star

points (of a star)

Part 1 — core work



This star was made with `Star(200, 200, 80, 5)`. What does the last number, 5, control?

Write a Line from the top-left corner (0, 0) to the point (250, 300).

PART 2 — keep going if you finish early.

An Oval uses a width and a height instead of one radius: `Oval(cx, cy, width, height)`. Write an Oval centered at (200, 200) that is 160 wide and 90 tall.

Name	Date	Period
Partner	Points ____ / ____	Marked by

Warm-up 4 — Colors: fill and border

START HERE

A	Pace check — circle one: () feeling good () a bit distracted () stuck — I could use help
B	☐ got it ☐ shaky ☐ not yet I can set a shape's fill color.
C	☐ got it ☐ shaky ☐ not yet I can add a border and set its width.
D	Looking back at last class — one thing that clicked, and one thing still fuzzy:
E	If you finish early today, what will you try next?

Vocabulary — write each in your own words

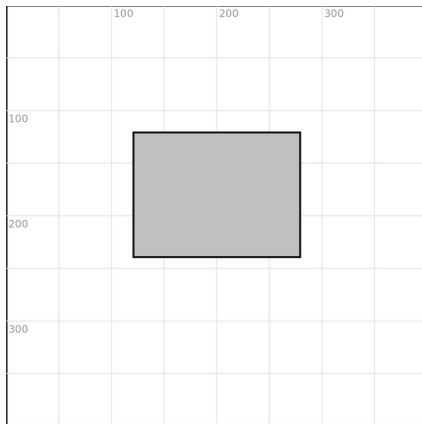
fill

border

borderWidth

color name

Part 1 — core work



The gray box below is filled gray. Write the Rect code that fills it gray AND adds a black border 4 pixels wide.

```
Rect(120, 120, 160, 160, fill='gray', border='black', borderWidth=4)
```

Name two different ways you could make this box look lighter.

PART 2 — keep going if you finish early.

Colors can be named ('crimson') or built with `rgb(r, g, b)`. Write an `rgb()` color you think would look orange.

Name	Date	Period
Partner	Points ____ / ____	Marked by

Warm-up 5 — Opacity and layering

START HERE

A	Pace check — circle one: () feeling good () a bit distracted () stuck — I could use help
B	☐ got it ☐ shaky ☐ not yet I can predict which shape is drawn on top.
C	☐ got it ☐ shaky ☐ not yet I can use opacity to see through a shape.
D	Looking back at last class — one thing that clicked, and one thing still fuzzy:
E	A goal — small is fine — for this class period:

Vocabulary — write each in your own words

layering (draw order)

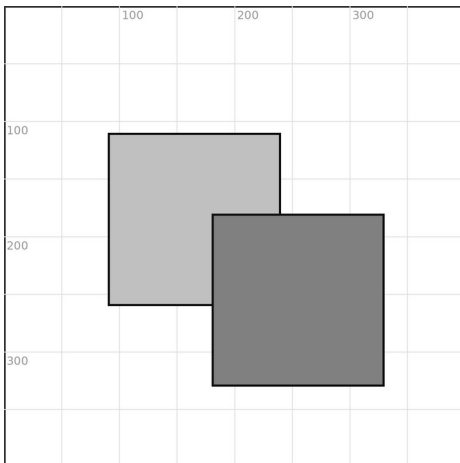
on top

opacity

overlap

Part 1 — core work

Two boxes overlap. The one listed **LAST** in the code is drawn on top.



Which box is on top — the lighter one or the darker one? How can you tell?

You want the **LIGHT** box on top instead. What do you change about the order of your two `Rect` lines?

PART 2 — keep going if you finish early.

opacity runs from 0 (invisible) to 100 (solid). What opacity would make a shape a faint 'ghost' you can still see through?

Name	Date	Period
Partner	Points ____ / ____	Marked by

Warm-up 6 — Labels and text

START HERE

A	Pace check — circle one: () feeling good () a bit distracted () stuck — I could use help
B	☐ got it ☐ shaky ☐ not yet I can put text on the canvas with Label.
C	☐ got it ☐ shaky ☐ not yet I can position and size a Label.
D	Looking back at last class — one thing that clicked, and one thing still fuzzy:
E	One word for how today is going so far:

Vocabulary — write each in your own words

Label

value (the text)

size (font)

align

Part 1 — core work

```
Label('Score: 0', 200, 40, size=20, fill='black')
```

In the Label above, which part is the text shown on screen? Which two numbers place it on the canvas?

Write a Label that shows your name, centered near the top of the canvas.

PART 2 — keep going if you finish early.

A Label is centered on its (x, y) by default. If your Label runs off the right edge, what could you change about its x?

Name	Date	Period
Partner	Points ____ / ____	Marked by

Warm-up 7 — Rotating shapes

START HERE

A	Pace check — circle one: () feeling good () a bit distracted () stuck — I could use help
B	☐ got it ☐ shaky ☐ not yet I can rotate a shape with rotateAngle.
C	☐ got it ☐ shaky ☐ not yet I can describe a rotation in degrees.
D	Looking back at last class — one thing that clicked, and one thing still fuzzy:
E	Something you are looking forward to this week:

Vocabulary — write each in your own words

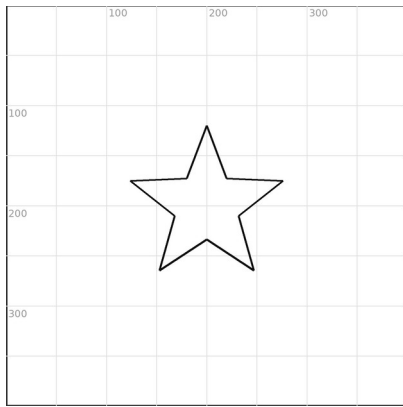
rotateAngle

degrees

clockwise

centerX / centerY

Part 1 — core work



`rotateAngle` turns a shape a number of degrees, clockwise. Add `rotateAngle` so the star below turns 36 degrees.

```
Star(200, 200, 80, 5, rotateAngle=36)
```

A full turn is 360 degrees. What `rotateAngle` brings a shape back to exactly where it started?

PART 2 — keep going if you finish early.

Rectangles rotate around their center too. In words, predict what `Rect(150, 150, 200, 40, rotateAngle=90)` would look like.

Name	Date	Period
Partner	Points ____ / ____	Marked by

Warm-up 8 — Putting it together: a scene

START HERE

A	Pace check — circle one: () feeling good () a bit distracted () stuck — I could use help
B	☐ got it ☐ shaky ☐ not yet I can read a multi-shape drawing and list its shapes.
C	☐ got it ☐ shaky ☐ not yet I can plan a small scene before I code it.
D	Looking back at last class — one thing that clicked, and one thing still fuzzy:
E	A person you could ask for help if you get stuck:

Vocabulary — write each in your own words

composition

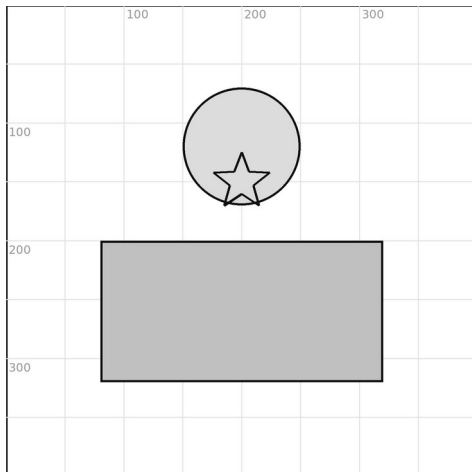
comment (#)

canvas

plan

Part 1 — core work

Read the scene below. List every shape you see, in the order you would draw them (back to front).



A comment starts with #. Write a one-line comment that explains the star in this scene.

PART 2 — keep going if you finish early.

Plan your own scene.

Sketch a simple scene — a house, a face, a robot — on the grid. Label at least three shapes with their function names (Circle, Rect, ...).

